

Jasper M. Dong

(925) 389-2017 | jasper.dong.2003@gmail.com | www.linkedin.com/in/jmudong

Education

University of Washington - Seattle

M.S. in Statistics - Advanced Methods & Data Analysis

Expected Graduation: March 2027

University of California - Davis

B.S. in Statistics - Statistical Data Science Track

Minor in Geographic Information Systems

Honors: Dean's Honor List (Fall 2021, Spring 2022, Fall 2022, Winter 2025)

Activities: Davis Data Science Club, HackDavis

2021 - 2025

GPA: 3.83

Experience

Graduate Data Science Intern - Lawrence Livermore National Laboratory

Present

Mentors: Ya Ju Fan, Jean-Paul Watson

- Analysis and visualization of gridded supply chain networks (~59k observations) for applications in agricultural resource transportation.
- Implemented Dijkstra's algorithm to build shortest path dictionaries using efficient data structures that reduced previous computational runtimes by ~92%.
- Developed a Haversine radius nearest-neighbor decision tree algorithm for accurate edge-weight predictions.

Statistics Tutor - University of Washington Department of Statistics

2025 - 2026

- Taught basic statistical concepts and intuition for undergraduate lower-division statistics courses in inference, computing, and applied methods.
- Provided motivating examples across a wide range of domain areas, including engineering, social sciences, and biological sciences.

Research Assistant - UC Davis Computational Climate and Ocean Group

2024 - 2025

Mentor: Maïke Sonnewald

- Performed exploratory data analysis and data preprocessing on large data sets (~1.2 million observations) using Python libraries and high performance computing tools.
- Combined the use of manifold learning and agglomerative clustering to identify unique ocean regions of interest based on different physical parameters.
- Implemented a point-based Chi-Square Test of Independence to ensemble clustering results for use as a clustering parameter validation method.

Projects

Agentic AI Workflow for Defect Detection and Simulation

2026

Lawrence Livermore National Laboratory

- Collaborated with team members to create scientific agentic workflows for materials science research using Model Context Protocol (MCP) tools.
- Utilized OpenAI Codex and GPT models for automation of analysis, reporting, and dashboarding of defects in 3D-printed octet lattice structures.
- Created downstream workflow applications using PyMAPDL and Ansys software for finite element analysis (FEA) simulations.

Comparing Forecasting Models for Univariate Time-Series

2026

University of Washington

- Processing of large weather datasets (~104k observations) consisting of 21 weather indicators for analysis of both short-term and long-term forecasting scenarios.
- Compared the accuracy and feasibility of traditional time-series prediction models (ARIMA, SARIMAX) against modern deep-learning models designed for time series forecasting (LSTM, Transformers).
- Implemented sLSTM architecture with exponential gating that shows an increase in accuracy (~5%) compared to vanilla LSTM architectures.

Skills

Programming Languages: Python, R, SQL, C, MATLAB, LaTeX, Git, HTML, CSS, JavaScript

Software: SLURM, Tableau, ArcGIS, Google Earth Engine, AWS

Spoken Languages: English, Chinese, Japanese, Dutch

Relevant Coursework: Linear Algebra, Data Structures, Discrete Mathematics, Applied Statistics, Probability Theory, Statistical Inference, Statistical Computing, Databases, Machine Learning, Optimization, Deep Learning, Software Development